



RAG assistant for Slovenian employment law

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Abstract

We built a question-answering assistant for Slovenian employment law using retrieval-augmented generation. We started from the COLESLAW legal corpus, ran into three systematic failures caused by outdated and low-authority documents, and fixed them by rebuilding the knowledge base from current official sources and iterating through five system prompt versions. The final system retrieves from PISRS statutes, government guidance, and related agencies, ranks sources by authority, and uses a strict Slovenian prompt that requires citations and refuses out-of-scope questions. On a 40-question evaluation set it achieves perfect retrieval with no false evidence, and the full RAG setup scores 2.95/5 for answer correctness and 1.00 for both citation support and refusal accuracy, compared to 0.50 and 0.00 for the baseline without retrieved context. The deployed assistant is available at <https://ai.koderverse.com/>.

Keywords

Slovenian employment law, legal question answering, retrieval-augmented generation, RAG, source grounding

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Motivation

Language models are not reliable for legal questions. They often sound confident while citing articles that do not exist, mixing up rules from different years, or applying foreign law to a Slovenian question. This is a real problem: employment law is one of the areas people most often want quick guidance on, and the questions have correct answers that can be verified against published statutes. We picked Slovenian employment law as our domain because it is well scoped, the official sources are publicly available online, and there are already a few Slovenian legal chatbots showing that people want this kind of tool [1, 2].

The assistant is informational rather than a legal advisor: it answers only Slovenian employment-law questions and refuses both out-of-scope requests and questions where retrieved documents are insufficient. This strict scope makes evaluation meaningful—we can test whether the system knows when not to answer, not just whether it answers correctly.

Related Work

The core idea we used is retrieval-augmented generation: instead of relying on what the model memorised during training, you retrieve relevant documents at query time and give them to the model as context [3]. For a legal assistant this is ideal because a claim grounded in a retrieved article can be checked

against the source, while a fluent answer from model memory alone cannot.

We also considered fine-tuning as an alternative. Techniques like LoRA can teach a model new output formats or styles without retraining the whole network [4], and tool-using approaches let models call external systems to get fresh data [5]. The trade-off between all these methods—prompting, RAG, and fine-tuning—has been surveyed recently in the context of injecting domain knowledge [6]. We went with RAG because legal answers need to be grounded in current statutes, and that requires the actual text to be present at generation time rather than baked into model weights.

Corpus and Sources

Our first approach was to filter COLESLAW 1.0 [7], a large collection of Slovenian legal documents, down to employment-law text and use that as the knowledge base. It gave short retrieved contexts (around 190 words per question on average) and we quickly found three concrete answer-quality failures: one question about fixed-term contract duration surfaced the wrong ZDR-1 article, another about who pays sick leave after day 30 returned outdated wording from an older version of the rule, and a third ambiguous question about leave entitlement produced a confident general answer instead of flagging the missing details. The underlying problem was that

COLESLAW is full of court decisions and older documents: there is plenty of text mentioning employment topics, but the current statutory text is not always the top result, and short context windows leave no room for corroborating sources.

To fix this we built a new knowledge base directly from the official online sources. We fetched and chunked documents from PISRS (the official Slovenian legal register), GOV.SI and MDDSZ (the main government and labour ministry guidance pages), ZZZS (health insurance), IRSD, eUprava, SPOT, ESS, and OPSI. Each chunk was tagged with a source type so the retriever could prefer higher-authority sources. We also kept a small set of COLESLAW court decisions as low-priority support for questions that specifically ask about legal interpretation or past court practice. The final corpus has 1,371 chunks in total: 1,059 primary-law chunks from PISRS statutes, 189 official interpretation chunks, 93 operational-guidance chunks, and 30 case-law chunks.

The authority order we enforced at both retrieval and prompt level is:

1. Statutes from PISRS, especially ZDR-1 (the Employment Relations Act) [8].
2. Official guidance from GOV.SI and MDDSZ [9, 10].
3. Operational guidance from ZZZS, eUprava, SPOT, ESS, and OPSI [11, 12].
4. Court decisions from sodnapraksa.si and COLESLAW, used only as supporting context.

Moving to this official corpus fixed all three earlier failures and grew average context length from around 190 to 1,430 words per question. We also built a source monitor that checks whether all 37 official sources are still reachable, so the corpus can be kept up to date as laws change.

System

For retrieval we use BM25, a standard lexical ranking algorithm, with a simple term-overlap fallback if the BM25 library is not available. We apply Slovenian-specific preprocessing: common stopwords are removed and a light rule-based stemmer handles the most frequent inflection patterns in the language.

On top of the raw BM25 score we apply two domain-specific adjustments. The first gives a score bonus to higher-authority source types: primary law gets the biggest boost, then official interpretation, then operational guidance. Case-law chunks receive a large negative penalty unless the question explicitly asks about court practice or legal interpretation, so that a question about the letter of the law does not accidentally surface a court case as the top result. The second adjustment handles actor-sensitive questions. Employment law questions are often asymmetric—what an employer can do and what an employee can do are different things—so we add small boosts when the retrieved text specifically matches the actor mentioned in the question.

The system prompt took five iterations: from a basic refuse-if-unsure rule, adding an internal context-check step, then vague-question handling, then source-priority instructions mirroring the retriever. The final version requires explicit citations on every non-refusal answer, mandates ambiguity explanations before answering vague questions, and requires all thresholds and tiers to be listed rather than just the first.

For the model itself we ran an arena evaluation across llama3, Mistral 7B, Gemma 3 4B, and two Qwen variants; every model scored 5.00/5 with RAG context on a smoke test—the questions were straightforward and the retrieved context directly contained the answers, so any capable model could synthesise a correct response—confirming prompt and retrieval quality dominate over model choice. We also evaluated GaMS-2B-Instruct [13], a Slovenian-focused model recommended by the course assistant; results are discussed in the Evaluation section. We chose Mistral 7B for deployment at <https://ai.koderverse.com/> (temperature 0) for its stronger citation discipline and overall correctness.

We prepared a fine-tuning dataset (176 training and 44 validation examples) but did not use it: retrieval quality was the bottleneck rather than output style, and a fine-tuned model's legal knowledge would go stale as statutes change.

Evaluation

We measured retrieval and answer quality separately: a retrieval failure and a generation failure produce the same wrong answer but need different fixes.

For retrieval we measured two things. Hit@3 checks whether the chunk containing the correct answer is among the top three results the retriever returns. The false-evidence rate checks the opposite failure mode: for questions that have no answer in the corpus, does the retriever still return a chunk that looks relevant? A high false-evidence rate would trick the model into generating a confident but unsupported answer for a question it should refuse.

For answer quality we used seven metrics. *Correctness* (0–5) measures how accurately the answer matches the expected legal answer—0 means completely wrong or invented, 5 means fully correct with all relevant details. *Grounding* (0–5) measures how well the answer stays within the retrieved documents rather than drawing on model memory; a well-grounded answer would not change if the model's training data changed. *Completeness* (0–5) measures how fully the answer covers the question, including thresholds, exceptions, and edge cases mentioned in the law. *Clarity* (0–5) measures how readable and well-structured the answer is, independent of whether it is correct. *Hallucination* (0–5) measures how much the model invents information not supported by the retrieved context—here 0 is best and 5 is worst, so this scale runs in the opposite direction from the others. *Citation support rate* (0–1) is the fraction of answers that include at least one verifiable reference to a specific article or official source. *Refusal accuracy* (0–1) is the fraction of deliberately unanswerable questions that the model correctly declines to answer.

rather than guessing.

The evaluation set has 40 questions across twelve topic areas—annual leave, fixed-term contracts, termination, sick leave, minimum wage, and deliberately ambiguous or out-of-scope questions—the last category being critical for a legal domain, so refusal accuracy is treated as a first-class metric.

Table 1. Retrieval results. Higher Hit@3 and lower false evidence are better. The official corpus was evaluated on the full 40-question set; the initial COLESLAW slice was measured on a 5-question smoke test covering only answerable questions. The main practical difference is context length.

Corpus	Hit@3	False ev.	Ctx. words
Official corpus	1.000	0.000	1430.8
Initial COLESLAW slice	1.000	0.000	189.9

Table 2. Answer quality on the 40-question set. Baseline rows have no retrieved context. All rows use the strict system prompt (strict_legal_rag_sl.v3) and temperature 0. GaMS-2B is the Slovenian GaMS-2B-Instruct model evaluated with the same corpus and prompt as Mistral. Correctness and hallucination are 0–5; citations and refusal are fractions.

Variant	Correct.	Halluc.	Citations	Refusal
Baseline (llama3)	0.50	4.12	0.00	0.00
Strict RAG (Mistral 7B)	2.95	1.68	1.00	1.00
GaMS-2B baseline	1.38	1.48	0.00	0.40
GaMS-2B strict RAG	2.38	1.93	0.13	1.00

Table 3. The three evaluation questions that failed with the initial COLESLAW corpus and how they were fixed.

Question	Fix
q009: fixed-term contract duration	ZDR-1 art. 55 retrieved first; two-year limit cited
q012: sick-pay payer after day 30	Current 30-working-day / 31st-day rule retrieved from both ZDR-1 and ZZZS
q014: ambiguous leave question	Ambiguity warning issued; ZDR-1 arts. 159 and 161 cited

For Mistral 7B the improvements from adding RAG are substantial across every metric. Grounding rises from 0.12 to 4.83, meaning the model shifts from answering almost entirely from memory to drawing almost entirely from the retrieved text. Citation support goes from 0.00 to 1.00, and refusal accuracy from 0.00 to 1.00: without context the model freely invents legal provisions and never refuses; with context and the strict prompt it cites sources and correctly turns away every question it cannot answer. Answer correctness rises from 0.50 to 2.95—retrieving the right article and synthesising a complete answer from several paragraphs remain separate challenges.

GaMS-2B-Instruct, evaluated with the same strict prompt, shows a different pattern: without context its correctness

(1.38) is almost three times the llama3 baseline due to Slovenian pretraining; with RAG context both correctness (2.38) and refusal accuracy (1.00) improve substantially, matching Mistral on refusal. The remaining gap is citations: 13% of GaMS RAG answers cite a source versus 100% for Mistral, and correctness remains lower (2.38 vs. 2.95).

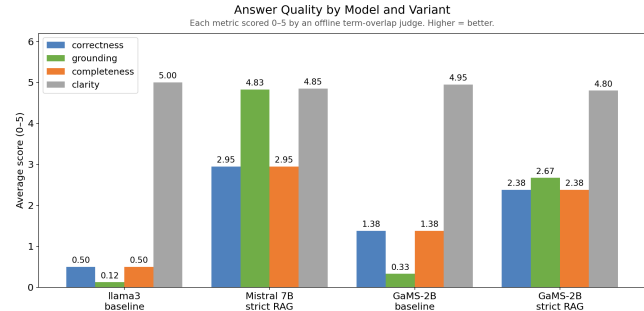


Figure 1. Correctness, grounding, completeness, and clarity (all 0–5, higher is better) for the four evaluated variants, all using the strict system prompt. Strict RAG (Mistral 7B) achieves the highest correctness (2.95) and grounding (4.83). GaMS-2B baseline (1.38) already outperforms the llama3 baseline (0.50) on correctness due to Slovenian pretraining; with strict RAG context correctness rises to 2.38, closing most of the gap with Mistral.

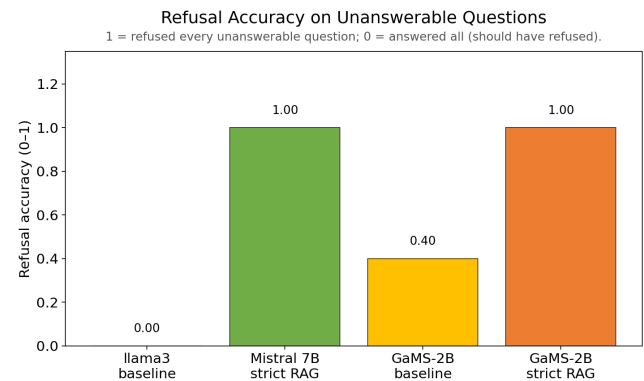


Figure 2. Refusal accuracy (fraction of unanswerable questions correctly declined) for all four variants. A score of 0.00 means the model answered every question it should have refused; 1.00 means it correctly declined all. With the strict system prompt, both Mistral strict RAG and GaMS-2B strict RAG achieve perfect refusal (1.00). The llama3 baseline (no strict prompt, no context) refuses nothing.

Reproducibility and Compliance

All results are reproducible: one make command rebuilds the corpus, reruns evaluations, regenerates charts, and compiles the report PDF. Corpus chunks, evaluation questions, and result snapshots are committed; secrets are never stored. The raw COLESLAW archive is omitted (too large) but derived chunks and conversion scripts are included.

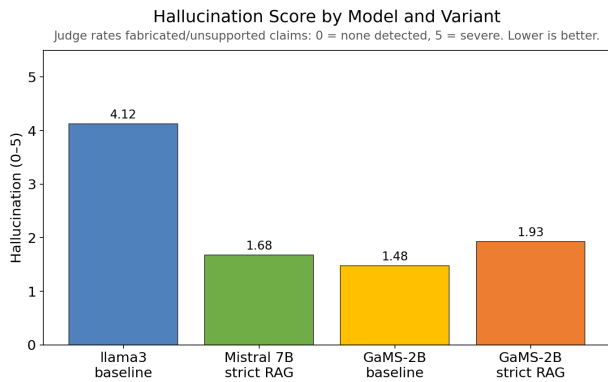


Figure 3. Hallucination score (0–5, lower is better) for all four evaluated variants (strict system prompt throughout). A score of 0 means no invented content; 5 means mostly fabricated. The llama3 baseline (no strict prompt, no context) scores highest at 4.12. Both Mistral strict RAG (1.68) and GaMS-2B strict RAG (1.93) achieve substantially lower hallucination with retrieved context. GaMS-2B baseline (1.48) is low because the strict prompt causes frequent refusals rather than generation.

Limitations and Future Work

The biggest gap is expert legal validation: automated scores (2.95/5 correctness) are not the same as a lawyer verifying answers are correct under Slovenian law. The system is informational only.

On the technical side, the retriever uses lexical matching, which means it can miss a question that asks about the same rule in different words. Dense retrieval or a learned reranker trained on employment-law question-document pairs would be the next improvement to try. We held off because it would require a GPU for training and the lexical retriever already achieves perfect Hit@3 on our evaluation set, so the gains would only show up on harder paraphrase cases we have not tested yet.

The main remaining gap for GaMS-2B is citation discipline: it already matches Mistral on refusal accuracy but includes source references in only 13% of RAG answers versus 100% for Mistral. Fine-tuning GaMS-2B specifically on citation behaviour—training it to output source references in the required format—would likely close most of the remaining gap while retaining the Slovenian language quality.

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